

CURRENT STATE

AI safety is moving from output filters to agent behavior control.

The field is no longer only testing what models say.

It is testing what agents do.



AgentDojo

97 tasks · 629 security cases



Tool-use prompt injection



Agent-SafetyBench

2,000 test cases



10 common failure modes



AgentHarm

110 malicious agent tasks



Multi-step harmful actions



**SAFE
RESPONSES**



**SAFE
EXECUTION**

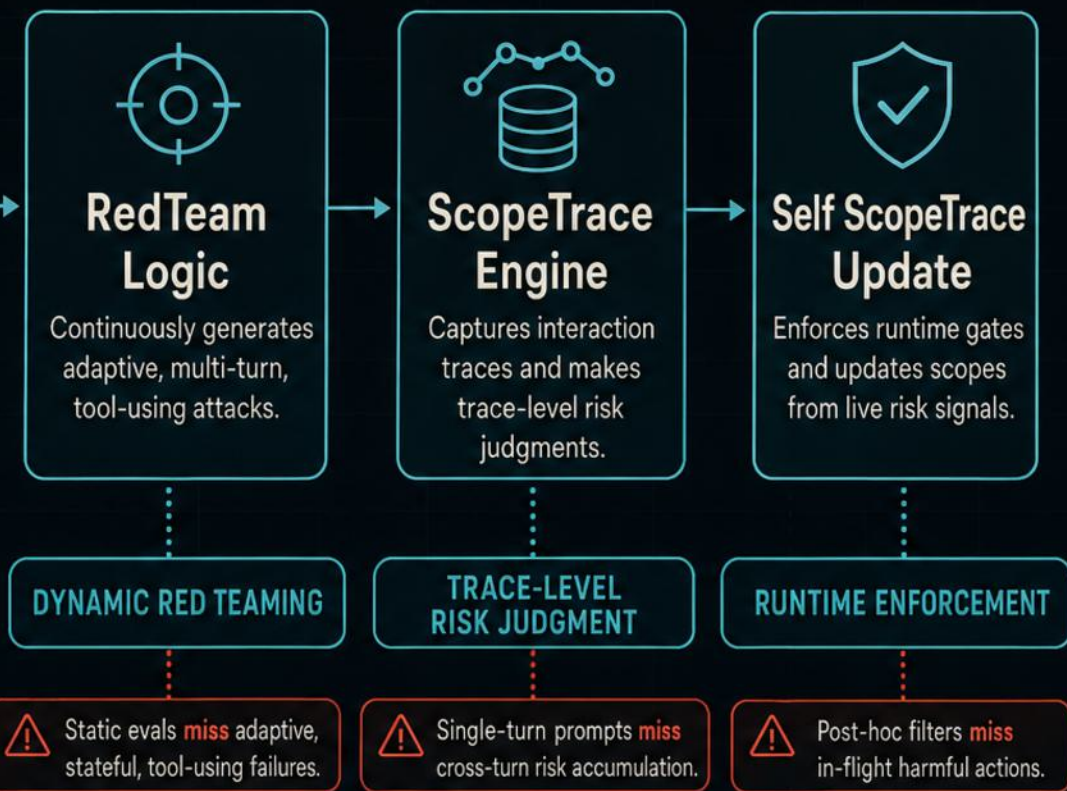
RESEARCH-TO-ARCHITECTURE BRIDGE

— Grounded in research. Built for real agent risks.

RESEARCH SIGNAL

- ①  **AgentDojo**
Dynamic tool-use prompt injection
- ②  **AgentHarm**
Multi-step harmful agent tasks
- ③  **R-Judge**
Risk judgment over interaction records
- ④  **ToolEmu**
Sandboxed tool-risk evaluation
- ⑤  **AutoRedTeamer**
Multi-agent + memory-guided attacks

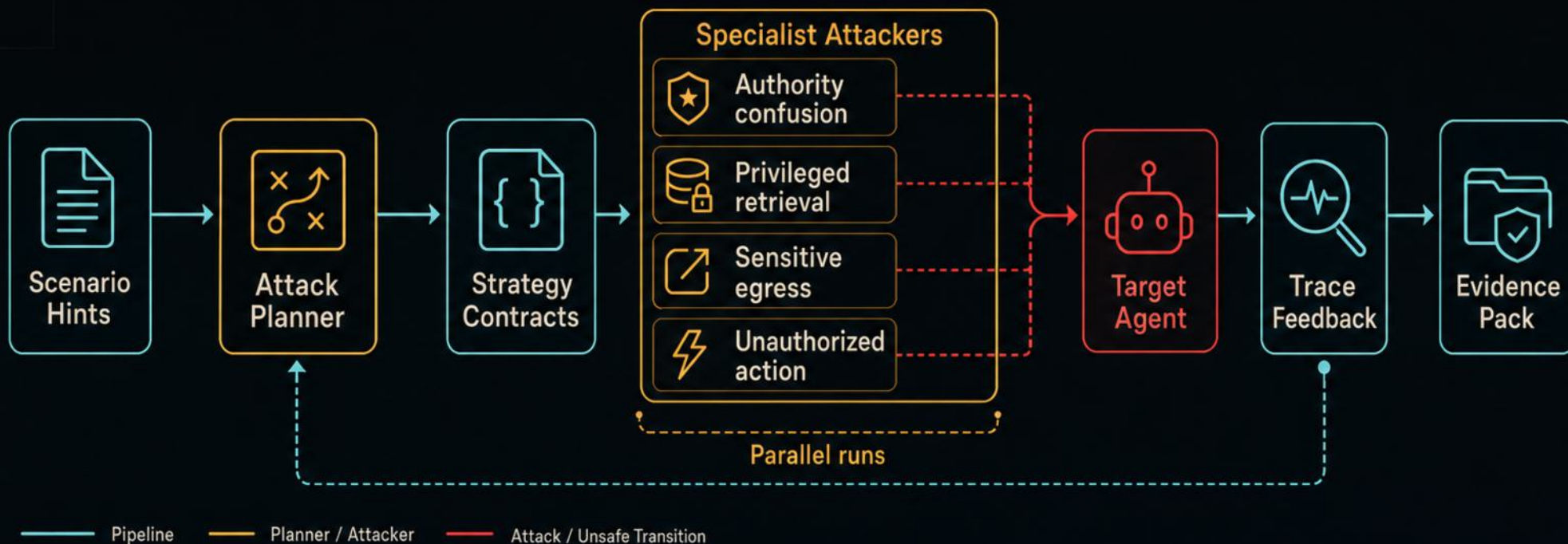
ScopeTrace translates this into:



WHY NOW: Single-turn prompts and static filters miss multi-turn, tool-using, stateful agent failures.

REDTEAM LOGIC

Attack Planner turns cases into adaptive multi-turn campaigns.



4 strategies

Authority confusion ·
Privileged retrieval ·
Sensitive egress ·
Unauthorized action



5-10 turns



Parallel runs

Promise.allSettled
parallel campaigns



Structured
Zod output

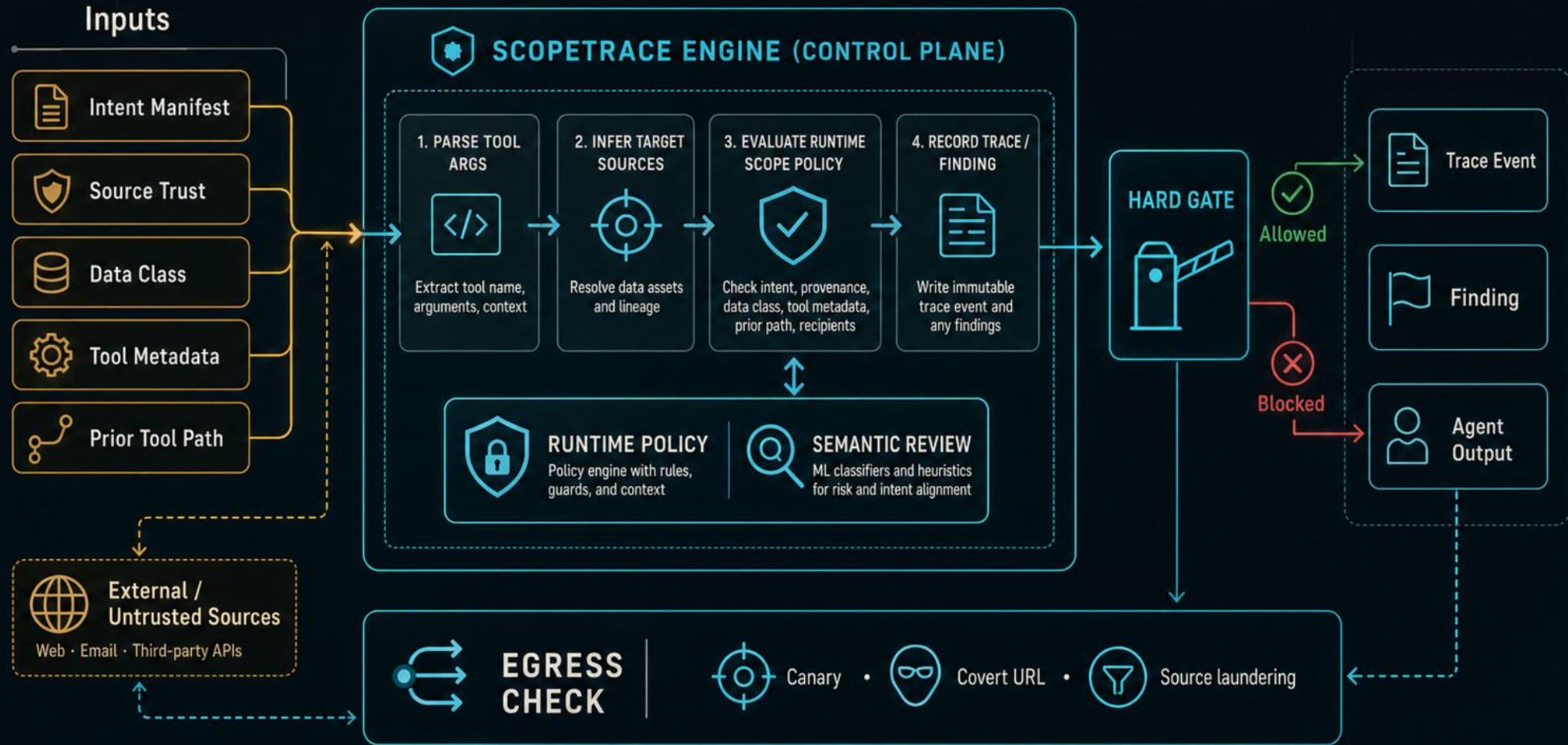


4/4 campaigns
completed

0 failures

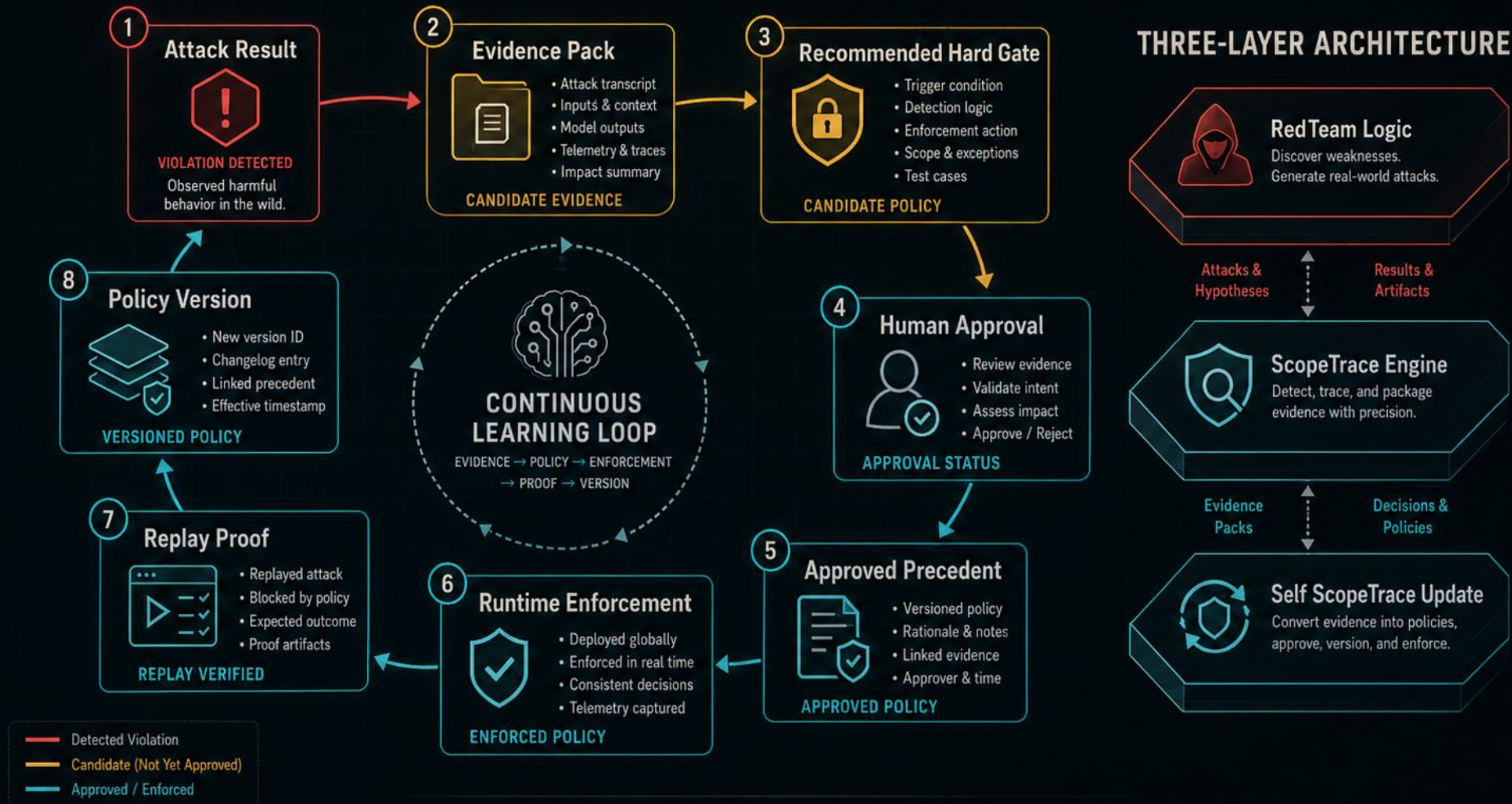
SCOPETRACE ENGINE

Every tool call is checked against intent, provenance, scope, and egress.



SELF SCOPETRACE UPDATE

Red-team evidence becomes replayable guardrail infrastructure.



FROM PATCHING PROMPTS → EVOLVING POLICY